

Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	
Project Name	LaunchPad AI – Smart Business Idea Evaluation System
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Proper indentation maintained throughout Python, HTML, CSS and JavaScript files.	Yes	Improves readability and code consistency.
2	Proper File Structure	Project organized into Templates, Static, Models, Routes, Services and Database folders.	Yes	Follows Flask project structure.
3	Meaningful Variable Names	Descriptive variable names are used for startup details, reports and AI responses.	Yes	Makes the code easier to understand.
4	Function / Method Names	Functions are named based on their purpose such as analyze_startup(), build_prompt(), history(), etc.	Yes	Improves code clarity and maintainability.
5	Code Comments	Important sections include comments explaining routes, AI integration and database operations.	Partial	Additional comments can be added for better documentation.
6	Modular Design	Frontend, backend, AI services and database logic are separated into different modules.	Yes	Makes the application scalable and reusable.
7	No Redundant Code	HTML templates, CSS and JavaScript are reused across multiple pages.	Yes	Reduces duplication of code.
8	Error Handling	Basic validation and exception handling are implemented for AI requests and user inputs.	Partial	Can be improved with more detailed exception handling.

Reusable Components / Modules

S.No	Component/Module Name	Language / Technology	Reusability Level
1	Flask Backend (app.py)	Python, Flask	High
2	Gemini AI Service	Python, Google Gemini API	High
3	Prompt Builder	Python	High
4	HTML Templates	HTML5, Jinja2	High
5	CSS Stylesheets	CSS3	High
6	JavaScript Files	JavaScript	High
7	SQLite Database	SQLite, SQLAlchemy	High
8	Flask Routes	Python	High
9	Report Dashboard	HTML, CSS, JavaScript	High
10	Images and Icons	PNG	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Project Structure	5	Organized using Flask architecture with separate folders for routes, services, templates and static files.
Readability	5	Uses meaningful variable names, proper formatting and consistent coding style.
Maintainability	5	Modular design allows individual components to be updated independently.
Reusability	5	AI services, templates, CSS, JavaScript and database modules can be reused in future projects.
Documentation	4	Basic comments are included; additional inline documentation can further improve clarity.
Error Handling	4	Basic validation exists, but more detailed exception handling can be implemented.
Overall Code Quality	5	The project follows clean coding practices with a modular, reusable and maintainable architecture.